



Supervised Learning - Classification

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LEARNING OBJECTIVES

When finished this intro and the notebook, learners should be able to:

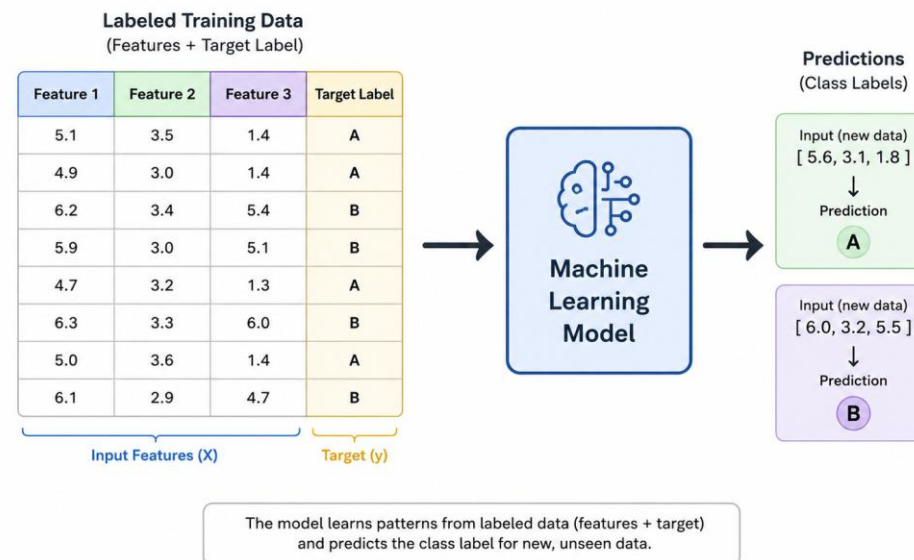
- Explain classification as supervised learning
 - Distinguish binary, multiclass, multilabel, and imbalanced classification
 - Describe the classification workflow
 - Compare common classifiers conceptually
 - Interpret accuracy, precision, recall, F1-score, confusion matrix, ROC/AUC
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WHAT MAKES LEARNING “SUPERVISED”?

- The model learns from examples with known answers
- Inputs are features
- Outputs are labels

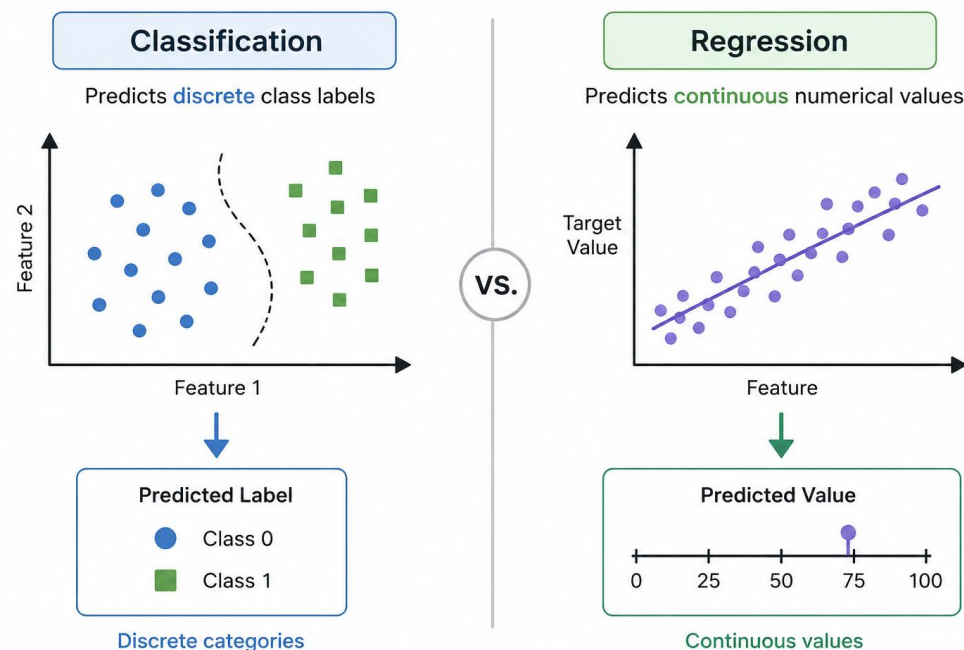
Goal: generalize to unseen cases





CLASSIFICATION VS REGRESSION

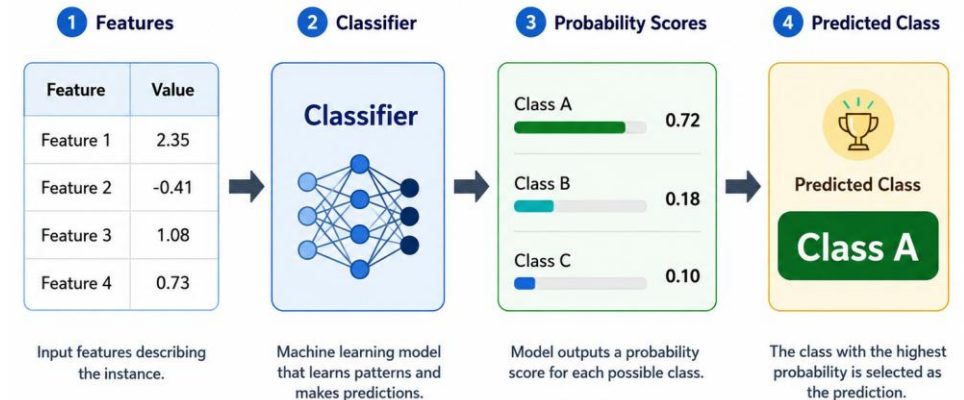
- **Classification:** predicts discrete classes
 - **Example:** spam / not spam
- **Regression:** predicts continuous values
 - **Example:** house price
- **Key question:** “Is the output a category or a number?”





ANATOMY OF A CLASSIFICATION PROBLEM

- **Features:** input variables
- **Classes:** possible categories
- **Model:** learned rule or boundary
- **Prediction:** selected class, often from probabilities





TYPES OF CLASSIFICATION

Binary: two classes

Multiclass: one class from many

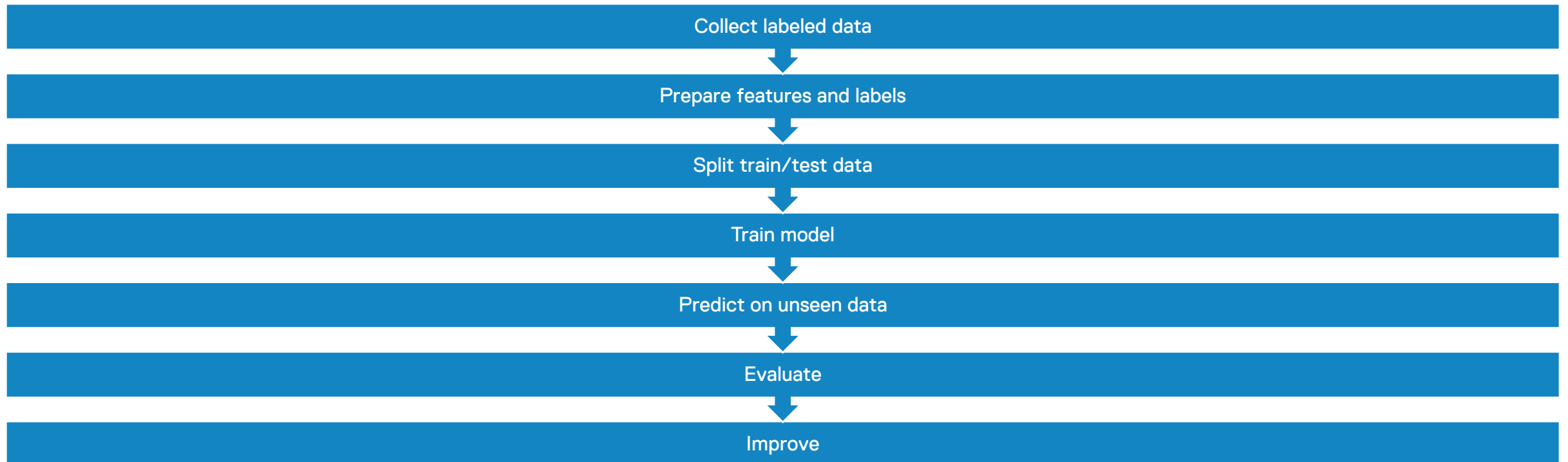
Multilabel: multiple classes can apply

Imbalanced: one class is much rarer than others





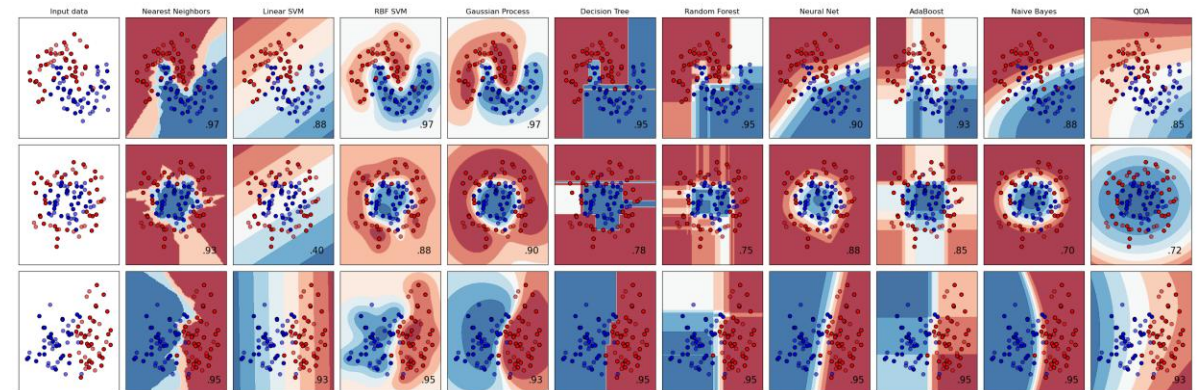
THE CLASSIFICATION WORKFLOW





DECISION BOUNDARIES

- Classifiers separate regions of feature space
- Simple models create simple boundaries
- Complex models can create flexible boundaries
- Good boundaries generalize; bad ones overfit



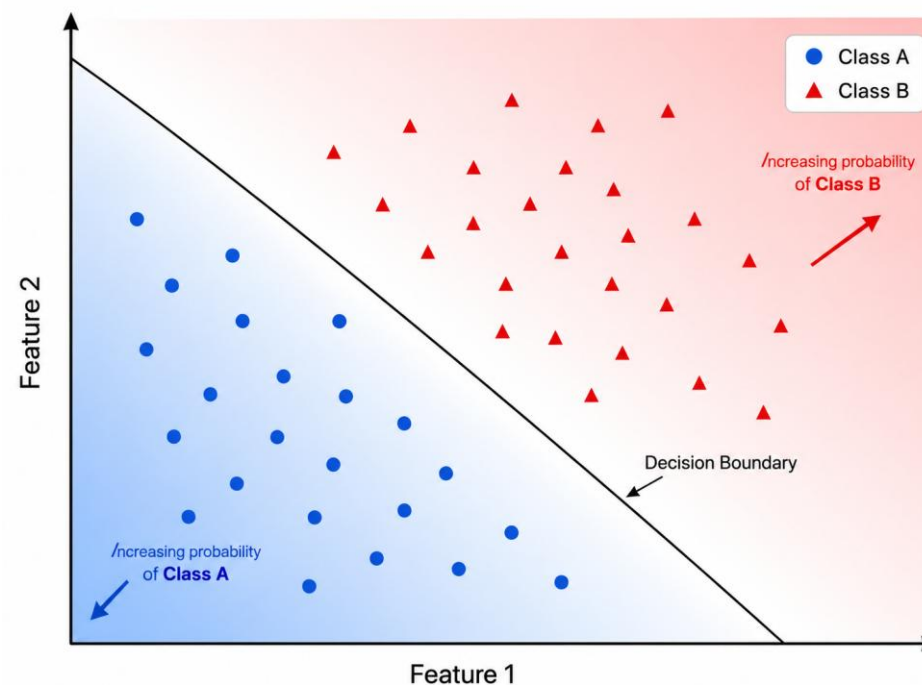


LOGISTIC REGRESSION

- A linear classifier despite the word “regression”
- Produces class probabilities
- Common baseline for binary classification

Strength: simple, fast, interpretable

Limitation: struggles with highly non-linear patterns

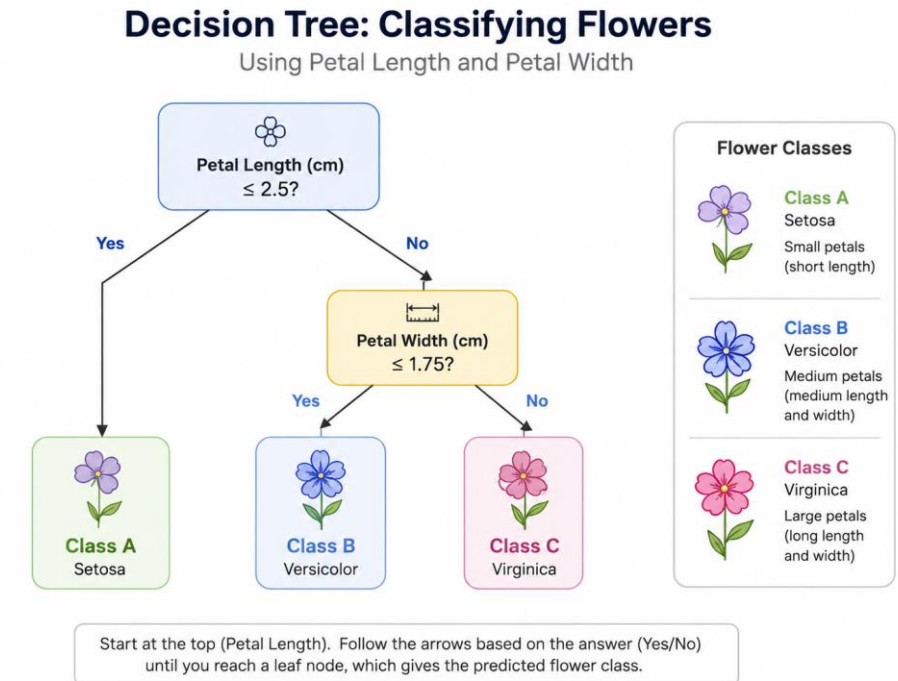




DECISION TREES

- Classify by asking feature-based questions
- Easy to explain as rules
- Can model non-linear patterns

Risk: overfitting if the tree grows too deep

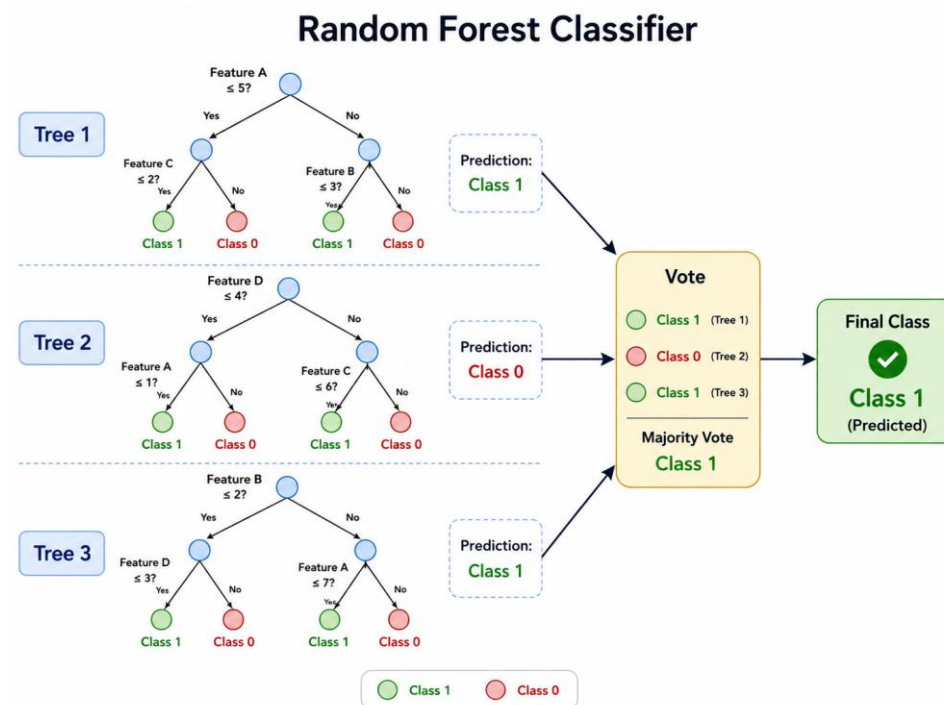




RANDOM FORESTS

- Ensemble of many decision trees
- Each tree votes for a class
- Usually more robust than a single tree

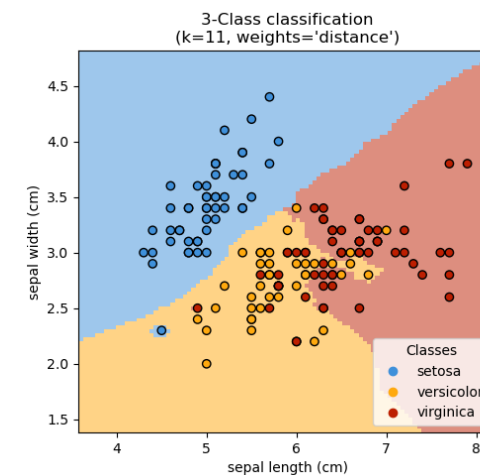
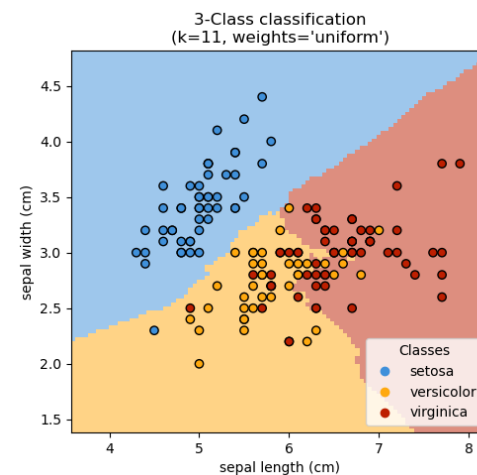
Tradeoff: less interpretable than one tree





K-NEAREST NEIGHBORS

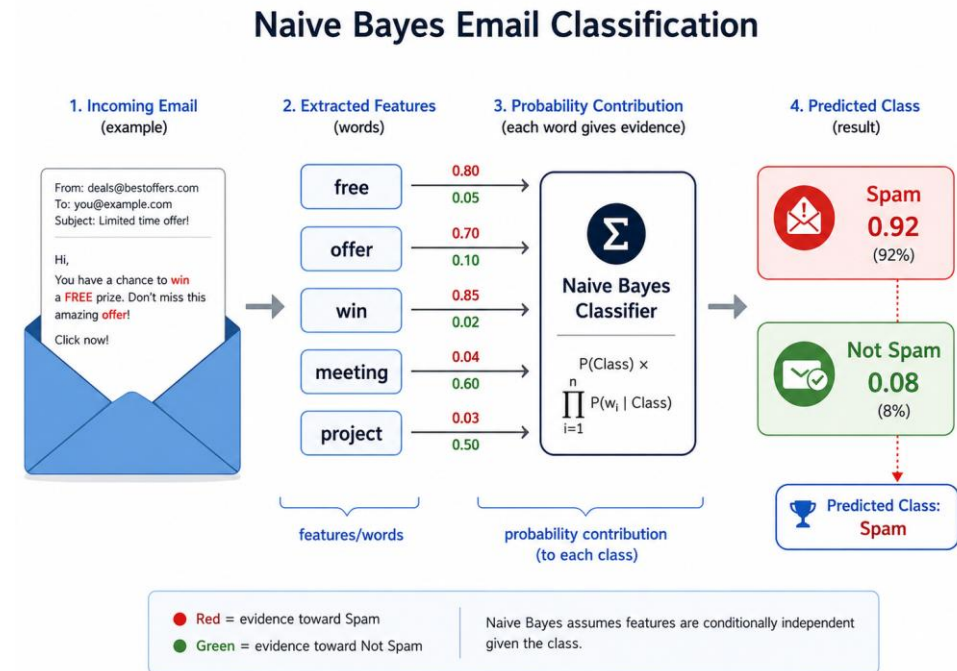
- Predicts based on nearby examples
- No strong training phase; stores examples
- Choice of k changes model behavior
- Works intuitively but can be slow on large datasets





NAIVE BAYES

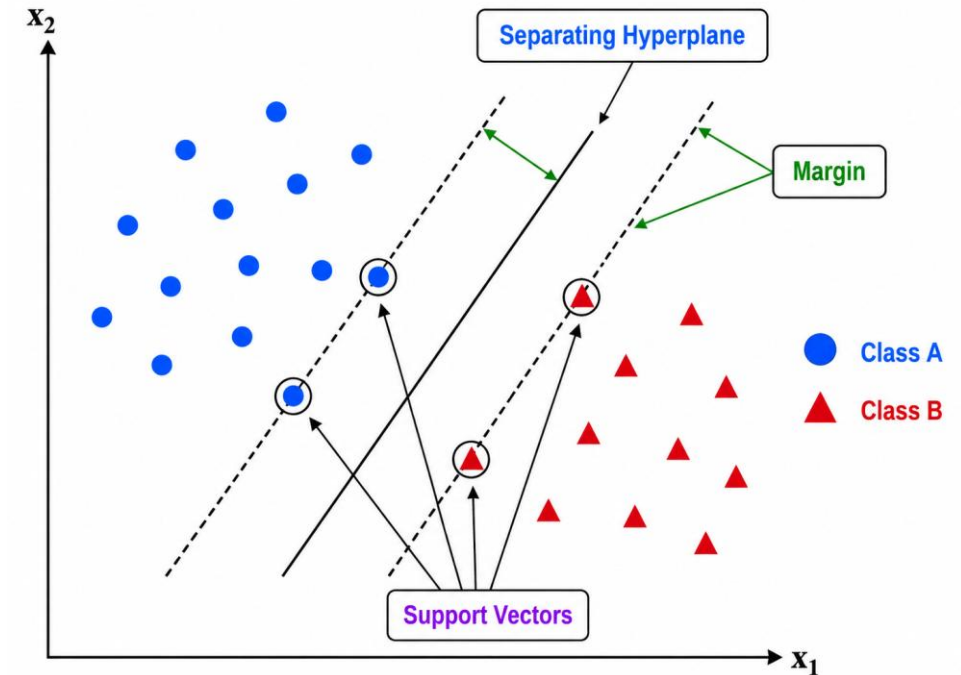
- Probabilistic classifier based on Bayes' theorem
- Assumes features are conditionally independent
- Often strong for text classification
- Fast and useful as a baseline





SUPPORT VECTOR MACHINES

- Finds a separating boundary with maximum margin
- Support vectors are key examples near the boundary
- Can use kernels for non-linear separation
- Powerful but less interpretable





CONFUSION MATRIX

- Shows correct and incorrect predictions
- Core terms: True Positive, True Negative, False Positive, False Negative
- Helps explain where the model fails

Confusion Matrix (2x2)

		Predicted	
		Predicted Positive	Predicted Negative
Actual	Actual Positive	TP true positive	FN false negative
	Actual Negative	FP false positive	TN true negative

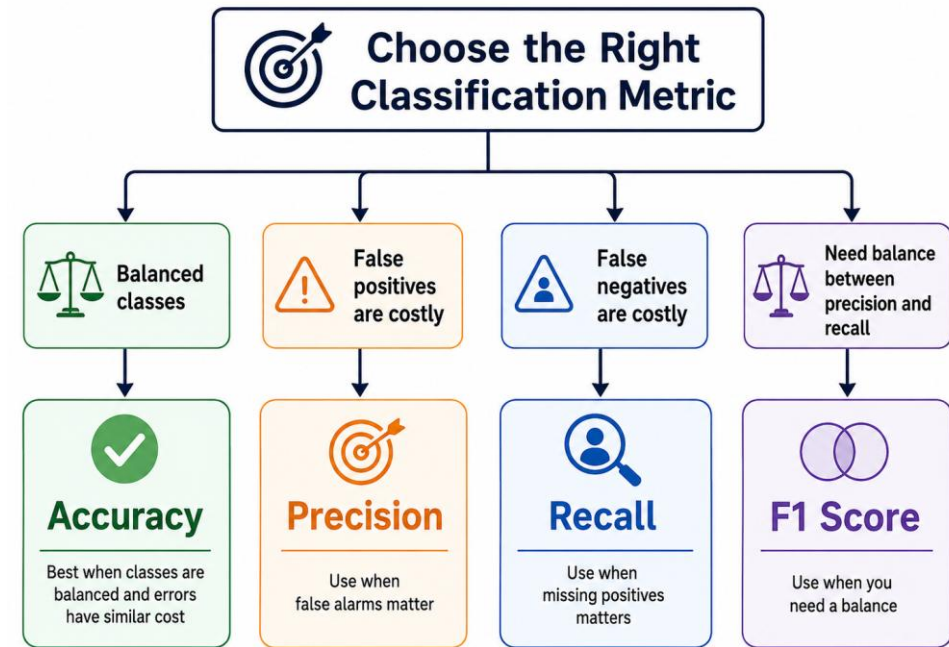
 Correct Predictions  Incorrect Predictions



ACCURACY, PRECISION, RECALL, F1

- **Accuracy:** overall correctness
- **Precision:** how trustworthy positive predictions are
- **Recall:** how many real positives were found
- **F1-score:** balance between precision and recall

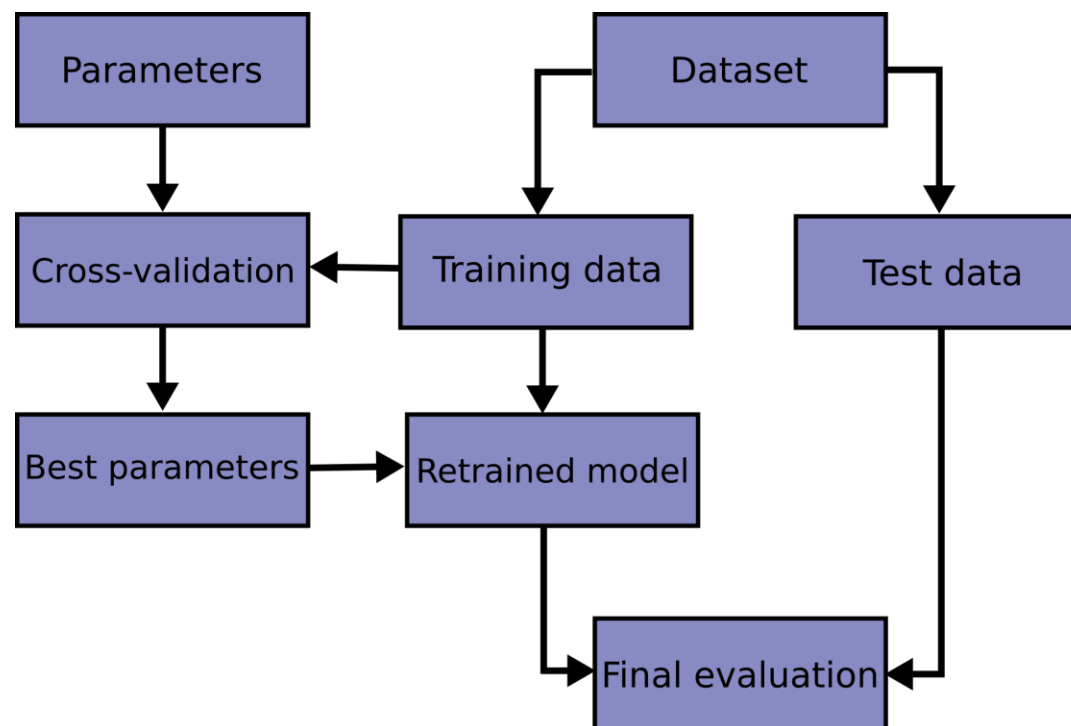
Metric choice depends on error cost





CROSS-VALIDATION AND GENERALIZATION

- One train/test split may be misleading
- Cross-validation evaluates across multiple splits
- Helps estimate real-world performance
- Useful for avoiding overconfidence





FINAL TAKEAWAYS

- Classification predicts categories from labeled examples
 - Algorithms differ in assumptions, flexibility, and interpretability
 - Evaluation is not just accuracy
 - The right metric depends on business and error costs
 - Good classification requires both modeling and judgment
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Technology Elevated